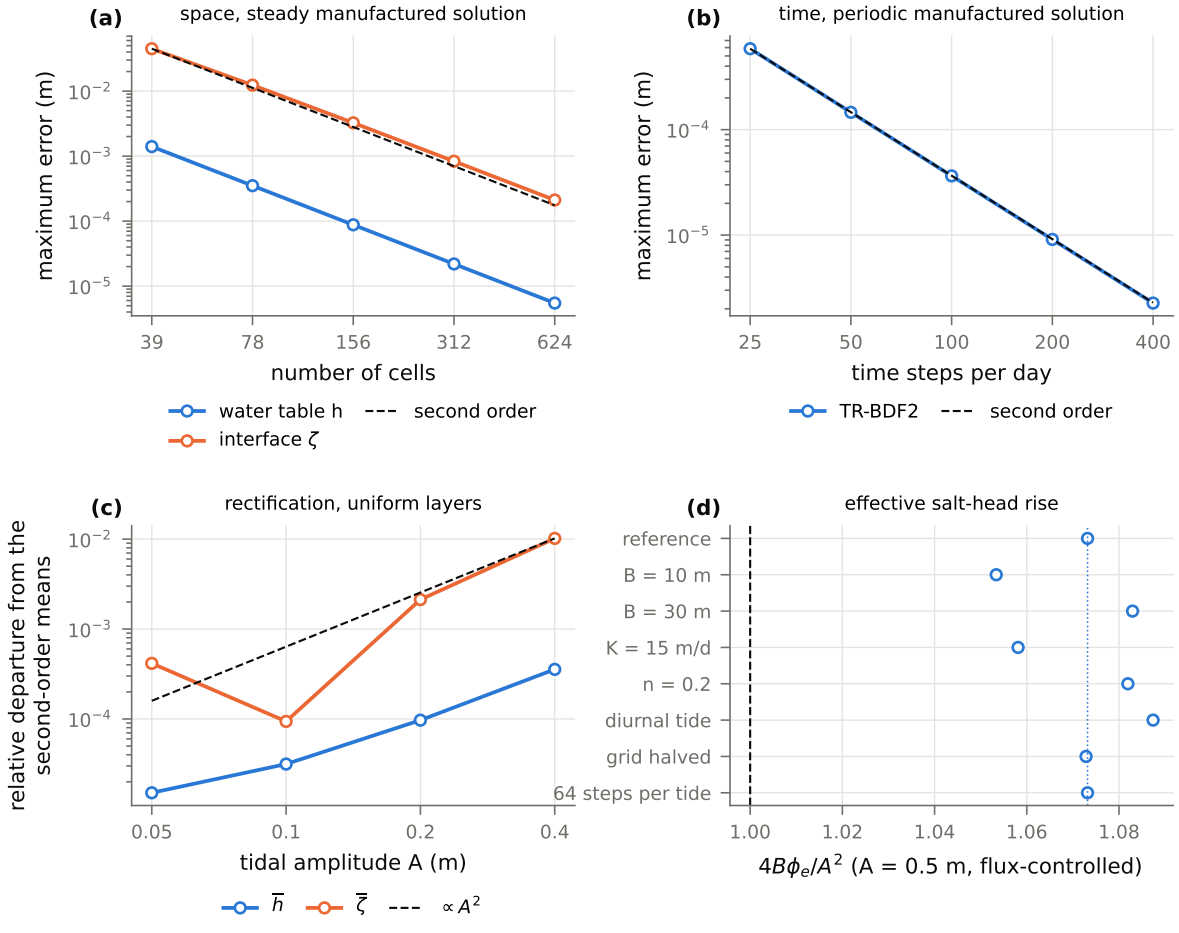
(a)
space, steady manufactured solution
maximum error (m)
10^{-2}
10^{-3}
10^{-4}
10^{-5}
39
78
156
312
624
number of cells
water table h
second order
interface ζ
(b)
time, periodic manufactured solution
maximum error (m)
10^{-4}
10^{-5}
25
50
100
200
400
time steps per day
TR-BDF2
second order
(c)
rectification, uniform layers
relative departure from the second-order means
10^{-2}
10^{-3}
10^{-4}
0.05
0.1
0.2
0.4
tidal amplitude A (m)
$\overline{h}$
$\overline{\zeta}$
$\propto A^2$
(d)
effective salt-head rise
reference
B = 10 m
B = 30 m
K = 15 m/d
n = 0.2
diurnal tide
grid halved
64 steps per tide
1.00
1.02
1.04
1.06
1.08
$4B\phi_e/A^2$ (A = 0.5 m, flux-controlled)